

ApexCloud REST API v3 & Enterprise Microservices Integration Guide

Technical Specification Document • API Version: 3.4.0 • Protocols: HTTPS / JSON / WebSockets / gRPC

1. Authentication, Tokens & Key Rotation

- 1.1 Bearer Token Authorization:** All REST requests must supply a valid API key in the Authorization header: Authorization: Bearer apex_live_sec_9941a.... Requests omitting this header fail immediately with HTTP 401 Unauthorized.
- 1.2 JWT Token Lifetime:** User session JWT tokens have a maximum validity window of **1,440 minutes (24 hours)**. Refresh tokens rotate every 7 days.
- 1.3 Key Rotation Policy:** Enterprise API keys must be rotated at least once every 90 days. During key rotation, ApexCloud supports a 48-hour grace period where both the old and new API keys are accepted simultaneously.

2. Rate Limits, Bursting & Concurrency Quotas

Rate limiting is enforced globally using a Redis token-bucket algorithm partitioned by API Key and Client IP:

Tier	Rate Limit (Req/Min)	Burst Allowance	Daily Max Limit	Concurrent WS Conns
Developer (Free)	60 req/min	10 requests	5,000 req/day	2 connections
Professional Tier	300 req/min	50 requests	50,000 req/day	10 connections
Enterprise Dedicated	2,000 req/min	250 requests	1,000,000 req/day	100 connections

When rate limits are exceeded, the server returns HTTP 429 Too Many Requests with headers: X-RateLimit-Limit, X-RateLimit-Remaining: 0, and Retry-After: .

3. Master Diagnostic Error Code Catalog

Error Code	HTTP Status	Root Cause Trigger	Corrective Action & Resolution
ERR-401-AUTH	401	Missing, expired, or revoked API Bearer token	Generate a new API key in ApexCloud Console.
ERR-403-TENANT	403	Cross-tenant resource access violation attempt	Verify resource ID belongs to your authenticated company_id.
ERR-422-SCHEMA	422	Pydantic payload schema validation failure	Inspect request JSON body against OpenAPI required fields.
ERR-904-SYNTAX	422	Malformed JSON syntax or invalid UTF-8 encoding	Ensure body is valid JSON with application/json header.
ERR-904-TIMEOUT	504	Upstream vector/LLM microservice timeout (>60s)	Retry with exponential backoff; reduce context top_k.
ERR-503-MAINT	503	Scheduled platform database maintenance window	Check https://status.apexcloud.io for live updates.
ERR-808-BUDGET	429	Tenant monthly token budget hard limit reached	Upgrade monthly budget cap in Usage & Budgets panel.

4. Idempotency & Safe Request Retries

For all state-mutating POST/PUT requests (e.g. /orders/create, /tickets/submit), clients must pass an Idempotency-Key: header. ApexCloud caches idempotency responses for **24 hours**. Retrying a request with the identical key returns the cached response without creating duplicate database records.

5. Webhooks & Asynchronous Event Subscriptions

ApexCloud delivers webhooks for events: order.created, order.delivered, ticket.escalated, payment.succeeded. Webhooks retry with exponential backoff (5 attempts over 2 hours). Clients must verify the HMAC-SHA256 signature in the X-Apex-Signature header using their webhook signing secret.